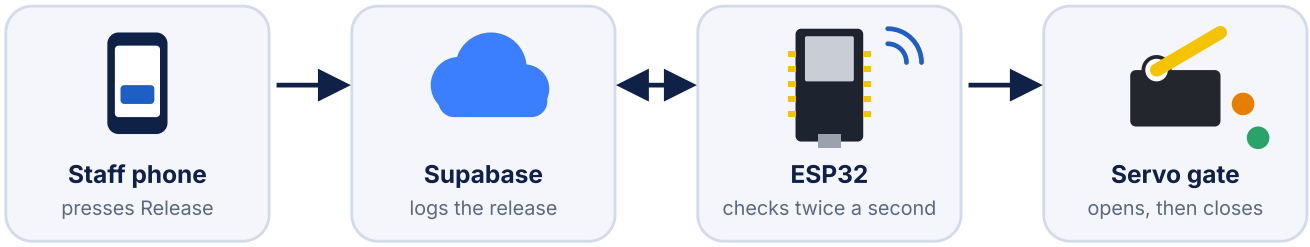
GENIUS ROBOTICS CHALLENGE
Resource Recovery Tournament · Sat 14 November 2026

BALL RELEASE SETUP GUIDE

LOLIN32 ESP32 board + servo, from a new board in its packaging to the first release from the staff page



From button to gate: about one second. No laptop or REV Control Hub needed at the field.

TIME NEEDED	SKILL LEVEL	WHO
About 2 hours, once	Beginner · a little soldering*	One person + the organiser

WHAT'S IN THIS GUIDE

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* Only if the board's two pin strips came loose in the bag (see Part 2).

PART 1

WHAT YOU NEED

Tick each item before you start

Gather everything first. Tick each box as you find it.

	Item	Notes
☐	**ESP32 board: WEMOS LOLIN32 V1.0.0** the new board, still in its packaging	Says **LOLIN32** on the board and **ESP-32** on the metal cover. This guide is written for this board. Other ESP32 boards need small changes, so ask first.
☐	**Micro-USB data cable**	Many cables only charge and can't carry data. A cable that came with a phone usually works.
☐	**Soldering iron and solder**, or someone who has one	Only if the board's two pin strips came loose in the bag (Part 2). Needed before wiring the servo, not for the earlier parts.
☐	**Windows PC or laptop** with internet	Only needed for setup (Parts 4–7 and 9). It doesn't need to be at the field.
☐	**Servo**	A normal hobby servo (MG996R, MG90S, SG90…) or a **REV Smart Robot Servo** in its normal servo mode (not continuous rotation). It comes with plastic arms ("horns") and small screws.
☐	**4 × AA battery holder** with wire leads, ideally with an on/off switch, plus **4 AA batteries** and a spare set	Powers the servo (about 6 V). For small SG90-type servos, use rechargeable NiMH AAs (4.8 V) or a 5 V supply. You could use a 5–6 V, 2 A+ adapter instead.
☐	**Small breadboard** (half size, 400 holes)	Joins the wires together without soldering.
☐	**Jumper wires**: 3 × male-to-male, 2 × female-to-male, 1 × female-to-female	"Female" ends slide onto the board's pins. "Male" ends push into the breadboard and into the servo plug. The female-to-female one is the **test wire** (Part 9). Buy a mixed pack.
☐	**USB power for the ESP32** at the field	A phone wall charger (best) or a fully charged power bank.
☐	**Wi-Fi with internet** at the venue	A **phone hotspot** is the most reliable. It must be **2.4 GHz** and must not need a login web page. Pick the phone now: it should be the one used on the day.
☐	**Website access**	A staff login for the staff page. Someone with access to the **Supabase dashboard** is needed once (Part 6).
☐	Small screwdriver, tape, cable ties, a small **plastic** box	For fitting the servo arm and mounting everything on the field (Part 11). Don't use a metal box, because it blocks Wi-Fi.

SUGGESTED TIMELINE

Thursday 12 Nov	Parts 1–7: unbox, install the software, put the program on the board (about 1 hour).
Thursday / Friday	Parts 8–10: wire the servo, set the angles, test from the staff page with real balls.
Friday 13 Nov	Part 11: mount it on the field and run a full rehearsal from the staff phones.
Saturday 14 Nov	Part 12: event-day checklist, 30 minutes before doors open.

PART 2

UNBOX AND POWER-ON TEST

10 minutes

> **Static electricity can damage the board.** Before you open the packaging, touch something metal and earthed, such as a metal tap or the metal case of a plugged-in PC. Work on a wooden or plastic table, not on carpet and not on a metal surface.

1. **Open the packaging.** The board usually comes in a silver or pink anti-static bag. It's sometimes pushed into a piece of black foam, and sometimes there are loose pin strips. Keep the bag so you can store the board in it later.
2. **Take the board out of the foam** if it's in some. Pull it out slowly and straight, rocking it gently from end to end, so the pins don't bend.

   > **Never plug in the board while its pins are in the black foam.** The foam conducts electricity and can short-circuit the pins.

3. **Hold the board by its edges.** Avoid touching the chips and the metal cover.
4. **Check the pins.** Look at the board from the side. Are there two rows of metal pins sticking out underneath, or only empty holes?

   > **Only empty holes, with two loose pin strips in the bag?** That's normal for the LOLIN32. Ask someone with a soldering iron to solder them on, with the long pins pointing **down** (away from the metal cover). You can do Parts 3–7 first, because they only use the USB socket. The pins are needed from Part 8.

FIRST POWER-ON TEST

This quick test shows the board and cable work before you install anything. Nothing is wired to the board yet.

1. **Plug the USB cable into the board**, then into the PC.
2. **Listen for the Windows "device connected" sound.** This is the main check.
3. **Look at the LEDs.** The LOLIN32 has no power light. Its small **CHG** LED may light up or flicker, which is normal when no battery is attached. The blue LED may blink if the board has a factory test program. Seeing neither is also fine.

> **The Windows sound played:** board and cable are fine. Carry on with Part 3.

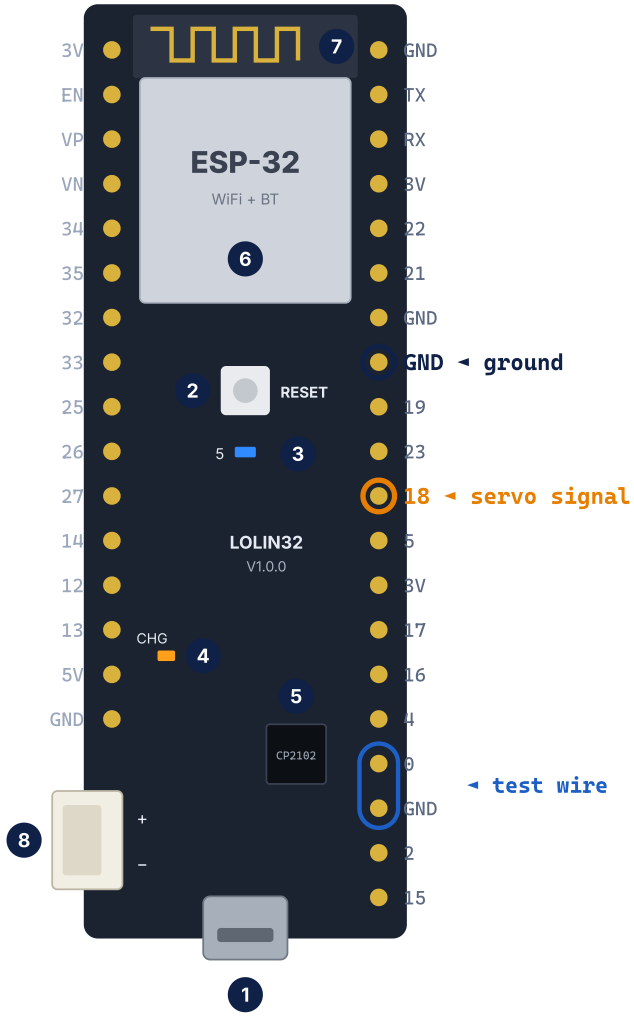
> **No sound:** try another USB socket on the PC, then another cable. The cable probably only charges, so swap it for a data cable before Part 5.

PART 3

GET TO KNOW THE BOARD

5 minutes

Find each numbered part on your own board, and fill in the box at the bottom right. You need these details later.



The **WEMOS LOLIN32 V1.0.0** from the top, with the USB socket at the bottom. Pins are labelled with plain numbers (**18**, not D18). The grey pins on the left aren't used in this project. Always go by the labels printed on your board.

1. **Micro-USB socket.** Power and programming.
2. **RESET button.** Restarts the board. The LOLIN32 has **no BOOT button**; the test wire does that job.
3. **Blue LED** (pin 5). Our program turns it on when the board is **connected to Supabase**.
4. **CHG LED.** Battery charging light. It may light or flicker when no battery is attached, which is normal.
5. **USB chip, CP2102.** It needs the Silicon Labs **CP210x driver** (Part 5).
6. **ESP-32 module** under the metal cover. This is the computer.
7. **Wi-Fi antenna.** Keep this end clear of metal.
8. **White battery plug.** Only for a 3.7 V LiPo battery. **Never plug the AA servo battery pack in here.**

P. **The pins you'll use**, all on the right: **18** (servo signal), a **GND** (ground), and **0** with the **GND** below it for the test wire.

CHECK AND WRITE DOWN

Pins soldered on? ☐ yes ☐ not yet

USB chip says: _____

Expected: CP2102. If it says CH340, tell whoever set this up.

PART 4

INSTALL THE ARDUINO SOFTWARE

20–40 minutes, once per PC
needs internet

A. INSTALL ARDUINO IDE 2

1. **On the PC, open** `arduino.cc/en/software` in a web browser.
2. **Download the Windows installer**, called `Windows · Win 10 and newer, 64 bits`. If a donation page appears, click `Just Download`.
3. **Run the downloaded file.** Click `I Agree`, keep the default options, click `Install`, then `Finish` with "Run Arduino IDE" ticked.
4. **Answer the first-start questions:**
   - Windows Firewall asks about network access: click `Allow`.
   - Windows asks "Would you like to install this device software?": click `Install`. These are USB drivers.
   - Arduino offers updates: it's safe to install or skip them. No extra libraries are needed for this project.

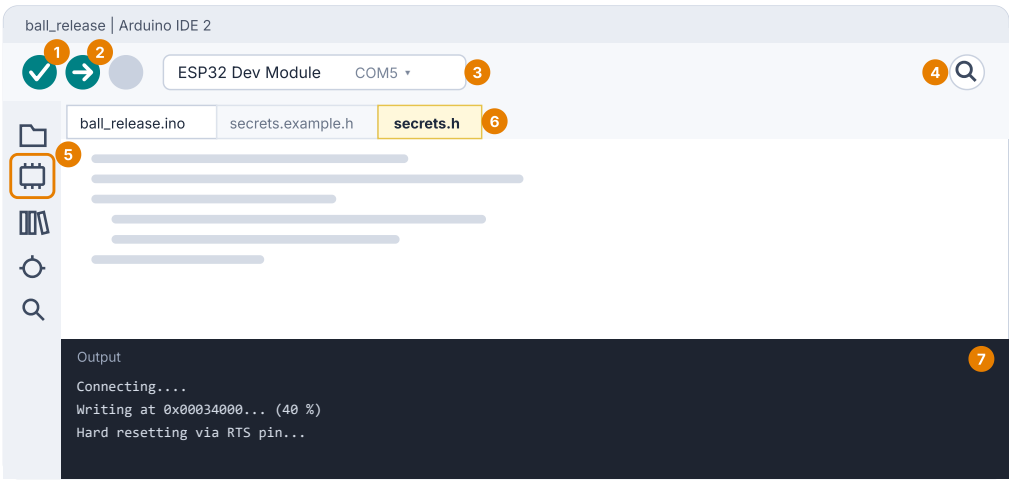
B. ADD ESP32 SUPPORT

1. **Open the Boards Manager:** click the **second icon** in the left sidebar (number 5 in the picture below), or use `Tools` › `Board` › `Boards Manager...`
2. **Type** `esp32` **in the search box.**
3. **Find "esp32 by Espressif Systems"** and click `Install`. The newest version is a **1.9 GB download**, so use internet with plenty of data, not a phone hotspot. On slow internet, pick **2.0.17** in the version list next to Install first (830 MB). The program is written to work with both.

> ! Pick **"esp32 by Espressif Systems"**, not "Arduino ESP32 Boards by Arduino", which is for a different board.

4. **Watch the Output panel** at the bottom. It should show `Downloading…` lines with progress. If nothing changes for 5 minutes, close Arduino completely, reopen it and click Install again. When it's finished, the label **"INSTALLED"** appears with a version number.

> i **"esp32 by Espressif Systems" isn't in the list?** Open `File` › `Preferences...`, paste this into **Additional boards manager URLs**, click `OK` and search again:
> `https://espressif.github.io/arduino-esp32/package_esp32_index.json`



Arduino IDE 2, roughly. **1** Verify (only checks the program) · **2** Upload (sends it to the board) · **3** Board and port selector · **4** Serial Monitor · **5** Boards Manager · **6** File tabs (secrets.h is where your details go) · **7** Output panel

PART 5

CONNECT THE BOARD TO THE PC

5–15 minutes

1. **Install the USB driver first.** The LOLIN32's USB chip is a **CP2102**, and Windows often doesn't install its driver by itself. Without it, the board won't get a COM port.
   1. On `silabs.com`, search for **CP210x USB to UART Bridge VCP Drivers**. On the **Downloads** tab, download **CP210x Universal Windows Driver**. It's a ZIP file.
   2. Right-click the ZIP and choose **Extract All**.
   3. In the extracted folder, right-click `silabser.inf` and choose **Install**. On Windows 11, click **Show more options** first. Accept the prompts.
2. **Plug the ESP32 into the PC** with the data cable. Plug it into the PC itself, not a USB hub, if you can.
3. **Check Device Manager:** right-click the Start button, click **Device Manager**, then open **Ports (COM & LPT)**. You should see **Silicon Labs CP210x USB to UART Bridge (COM…)**. The COM number in brackets is your board.
4. **Choose the board and port in Arduino:** click the board selector at the top (3), then **Select other board and port…**. Type `ESP32 Dev Module` in the left box and click it (**WEMOS LOLIN32** also works), click your COM port on the right, then **OK**.

My ESP32's COM port: _____

STILL NO COM PORT?

In Device Manager you see	Do this
Other devices › **CP2102 USB to UART Bridge Controller**, with a yellow warning sign	The driver isn't installed. Repeat step 1, making sure you chose **Install** on `silabser.inf`. Then unplug the board and plug it back in.
Nothing new appears, and Windows makes no sound	The cable probably only charges. Try another cable, and another USB socket on the PC.
A **USB-SERIAL CH340** entry, on a different board	That board uses a CH340 chip. On `wch-ic.com`, search for **CH341SER**, then download and run `CH341SER.EXE` and click **INSTALL**.

✓ **Done when:** the top bar in Arduino shows **ESP32 Dev Module** and your **COM** port.

PART 6

PREPARE THE DATABASE AND WEBSITE

5 minutes, once done by the organiser

This creates the "release" log and a secret **device token**. The token is the ESP32's password for the database. It needs someone who can sign in to the Supabase dashboard.

1. **Open** `supabase.com`, sign in and open the tournament project.
2. **In the left menu, click** SQL Editor, then New query.
3. **On the organiser's PC, open** `GRC Tournament\supabase\field_release.sql` in Notepad. Select everything with Ctrl+A, copy it with Ctrl+C, and paste it into the SQL Editor.
4. **Click** Run. The result at the bottom shows one row, headed **DEVICE_TOKEN (copy into secrets.h)**.
5. **Copy that long code** and paste it into Notepad for Part 7.

> ! **Treat the token like a password.** Don't print it, post it in a group chat or put it on the website. If you lose it, run the same file again and it shows the same token. To make a new token, see Part 13.

6. **Check the website:** sign in to the staff page and open the Score tab. Under the match clock there's a **Ball release** panel.

The Ball release panel says	Meaning
• Not set up yet	Steps 1–4 above haven't been done, or the page needs a reload.
• Dropper offline	Correct for now. The ESP32 connects in Part 7.
No Ball release panel at all	The website update isn't online yet. Ask whoever manages the GitHub repository to push it.

PART 7

PUT THE PROGRAM ON THE ESP32

15 minutes

A. GET THE PROGRAM FILES

- **On the organiser's PC:** they're in `Desktop\GRC Tournament\esp32\ball_release\`
- **On another PC:** open the GitHub repository, click **Code** › **Download ZIP**, then right-click the ZIP and choose **Extract All**. Open `GRC Tournament\esp32\ball_release\`

Keep the folder name `ball_release`. Arduino needs the folder to have the same name as the `.ino` file inside it.

B. OPEN IT AND FILL IN YOUR DETAILS

1. **Double-click** `ball_release.ino`. It opens in Arduino. You can also use **File** › **Open...**
2. **Click the** **secrets.h** **tab** (6) and fill in the four lines between the quote marks:

```
// The Wi-Fi it will use at the venue (2.4 GHz)
#define WIFI_SSID       "GRC-Hotspot"          // exact name, capitals matter
#define WIFI_PASSWORD   "your-password-here"

// Optional backup network, e.g. a second phone's hotspot ("" if none)
#define WIFI_SSID_2     ""
#define WIFI_PASSWORD_2 ""

// The token from Part 6
#define DEVICE_TOKEN    "3f9a1c…the long code…"
```

Keep the quote marks `"` and change only the text between them.

> **No secrets.h tab?** Click the ··· button at the right end of the tabs, then **New Tab**, and name it `secrets.h`. Copy everything from the **secrets.example.h** tab into it, then fill it in.

3. **Save** with **Ctrl** + **S**.

> **Make the hotspot 2.4 GHz**, because the ESP32 can't see 5 GHz Wi-Fi:
>
> - **iPhone:** go to **Settings** › **Personal Hotspot** and turn on **Maximise Compatibility**.
> - **Android:** in the Hotspot settings, set the **AP band** to **2.4 GHz**. Also turn off "Turn off hotspot automatically" if your phone has it.

> **Never upload secrets.h to GitHub**, especially through the GitHub website. It holds your Wi-Fi password and the token.

C. UPLOAD

1. **Check the top bar** (3) shows **ESP32 Dev Module** and your COM port.
2. **Click Upload** (2, the → arrow). The first time, "Compiling sketch..." takes 1–3 minutes.

3. **Watch the Output panel** (7). It shows `Connecting....`, then `Writing at … (10 %) … (100 %)`, and finally `Hard resetting via RTS pin...`. That last line means it's done.

> **Stuck on `Connecting…….` , or "Failed to connect"?** The LOLIN32 has no BOOT button, so use the test wire instead. Connect pin **0** to the **GND** pin below it, press **RESET**, click Upload again, and remove the wire once the percentages start. Press **RESET** when it's finished.

> **Red error text while "Compiling"?** Take a photo or copy the first red lines and send them to whoever set up the website. The most common cause is a missing quote mark `"` in secrets.h.

D. CHECK THAT IT WORKS

4. **Open the Serial Monitor** (4, top right). In its bar, set the speed to **115200 baud**.
5. **Press RESET** on the board once to restart it. Within about 10 seconds you should see:

```
GRC ball release starting
Connecting to Wi-Fi "GRC-Hotspot"...
Wi-Fi connected to "GRC-Hotspot", signal -52 dBm
Connected to Supabase. Waiting for the Release button (last release was #0).
```

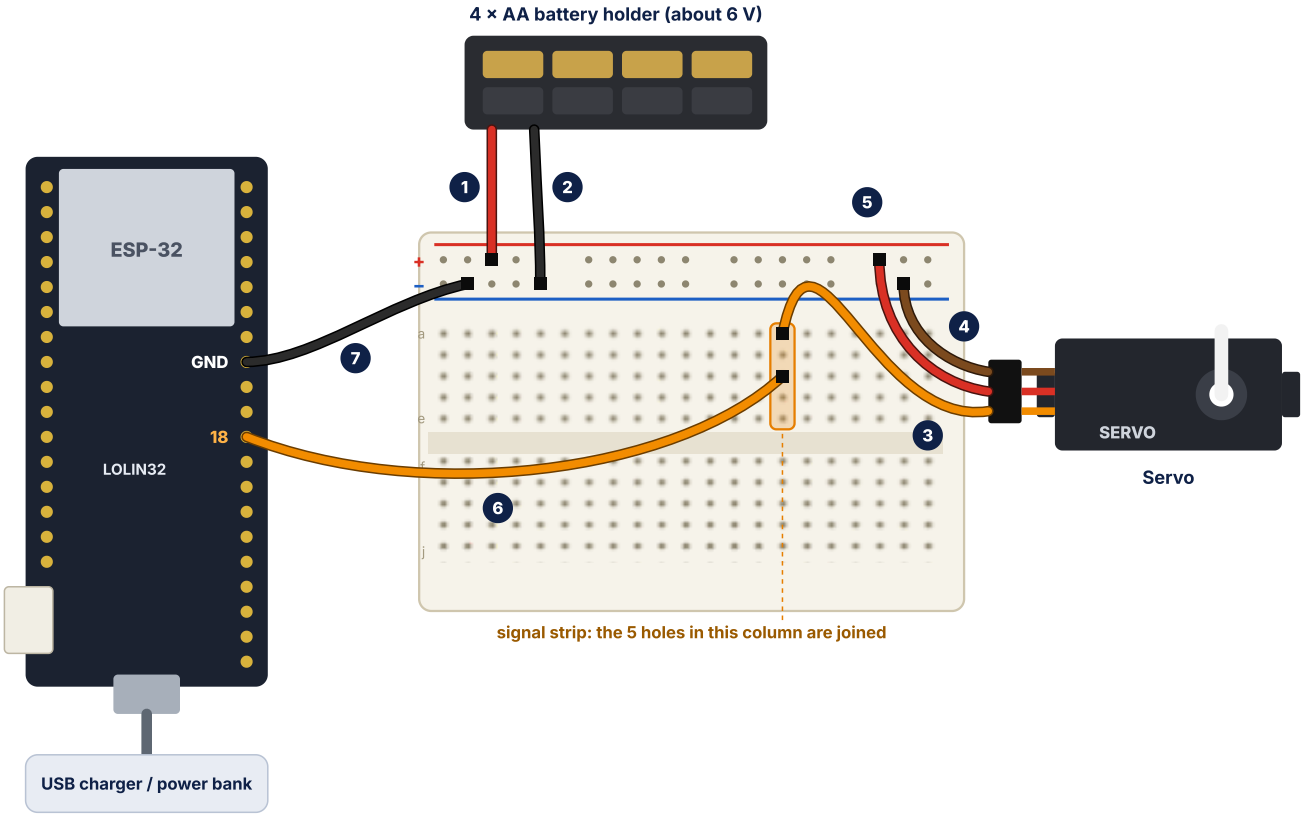
> **Done when:** the **blue LED** next to RESET **stays on**, and the staff page's Ball release panel shows **• Dropper online**. If the LED does the opposite (on while offline), change `LED_ACTIVE_LOW` near the top of `ball_release.ino` and upload again.

PART 8

WIRE UP THE SERVO

15 minutes

Before you start: the board's pin strips must be soldered on (Part 2). Unplug the ESP32's USB cable and take one battery out of the holder, or switch it off. Wire everything with the power off, check it against the table, then switch on. Pins **18** and **GND** are both on the right-hand side of the LOLIN32 (Part 3).



Jumper wire colours don't matter. The colours in the drawing just make it easier to follow. The wire **ends** (black squares) show exactly which hole each one goes in.

#	From	To	Wire
1	Battery holder **red** lead (+)	Breadboard **+ rail**	The holder's own lead
2	Battery holder **black** lead (–)	Breadboard **– rail**	The holder's own lead
3	Servo plug **red** (+)	Breadboard **+ rail**	Male-to-male
4	Servo plug **brown/black** (–)	Breadboard **– rail**	Male-to-male
5	Servo plug **orange/yellow/white** (signal)	A hole in row **a** of any column	Male-to-male
6	ESP32 pin **18**	The **same column** as wire 5 (rows a–e)	Female-to-male
7	ESP32 pin **GND**	Breadboard **– rail**	Female-to-male
8	ESP32 USB socket	USB charger or power bank	USB data cable

WIRING TIPS AND CHECKS

SERVO WIRE COLOURS

Brown or **black**	Ground (–)
Red (middle wire)	Power (+)
Orange, **yellow** or **white**	Signal

REV Smart Robot Servo: black = –, red = +, white = signal.

HOW A BREADBOARD JOINS WIRES

- The long **+** and **–** rows along the edge (the **rails**) are each joined all the way along.
- On some breadboards the rails are **split in the middle**: look for a gap in the red and blue lines. If yours has a gap, keep all the + and – wires on the same half.
- Each short column of **5 holes** (rows a–e) is joined. The groove in the middle separates a–e from f–j.

PRACTICAL TIPS

- **Servo plug:** push one end of each male-to-male jumper into the three holes of the servo's plug.
- **Battery leads** are soft, stranded wire. Twist the bare ends tightly so they push firmly into the holes. If they're too short or floppy, ask someone to solder on header pins, or use a holder with a plug.
- **A holder with an on/off switch** gives you a servo power switch, which is handy for setup and at the field.
- **The ESP32 doesn't go in the breadboard.** It's too wide. Let it sit beside the breadboard on the jumper wires.

Never connect the battery or the servo's red wire to any ESP32 pin, including 3V and 5V, **or to the board's white battery plug**. The ESP32 gets its power only from USB. The two meet only at **GND**.

Wire 7 (GND to GND) is required. Without it the servo twitches randomly or doesn't move at all.

CHECK BEFORE SWITCHING ON

- [ ] Battery red and servo red are both in the **+** rail, and nothing else is.
- [ ] Battery black, servo brown and ESP32 **GND** are all in the **–** rail.
- [ ] Servo signal and ESP32 pin **18** are in the **same 5-hole column**, on the same side of the groove.
- [ ] No bare wire ends are touching each other.
- [ ] Nothing is connected to the ESP32's **3V** or **5V** pins, or to its white battery plug.

All ticked? Put the battery back in, or switch the holder on, then plug in the ESP32's USB. The servo moves once to its **closed** position. That's normal.

PART 9

FIT THE SERVO AND SET THE ANGLES

15–30 minutes
needs the PC for changes

Keep fingers clear of the gate whenever the servo has power. It can move suddenly, for example when the board restarts.



CLOSED
`CLOSED_ANGLE = 0`
arm blocks the opening

OPEN
`OPEN_ANGLE = 90`
arm swings away, balls drop

One example layout. Your gate can look different: what matters is the arm position for "closed" and for "open".

1. **Switch on the servo battery, then plug the ESP32 into the PC** (the PC is needed to change the angles). The servo turns to the **closed** position and stays there.
2. **Fit the arm.** If it's already on, unscrew it and pull it off. Push the arm onto the shaft at the angle where it **shuts the gate**, then put the small screw back in.
3. **Test with the test wire.** Push one end of the female-to-female jumper onto pin **0** and leave it there. To test, push the other end onto the **GND** pin just below pin 0 for a second, then pull it off. The gate should open for 1.5 seconds and close again. The Serial Monitor shows `Opening the gate (test on pin 0)`. This works even without Wi-Fi.

   Don't leave pin 0 connected to GND while plugging in or pressing RESET. That starts the board in programming mode, so it waits instead of running. Pull the wire off GND and press RESET to fix it.

4. **Adjust if needed.** On the PC, open `ball_release.ino`. Change these lines near the top, then **Upload** again (Part 7C) and repeat step 3:

```
const int CLOSED_ANGLE = 0;                // gate shut (0 to 180)
const int OPEN_ANGLE = 90;                 // gate open (0 to 180)
const unsigned long HOLD_OPEN_MS = 1500;   // how long it stays open (1500 = 1.5 s)
```

What you see	Change
Opens the wrong way	Swap the numbers, for example `CLOSED_ANGLE = 90` and `OPEN_ANGLE = 0`.
Doesn't open far enough	Move `OPEN_ANGLE` further from `CLOSED_ANGLE`, for example 90 → 110.
Servo buzzes or strains when closed	It's pushing against something. Move `CLOSED_ANGLE` 5–10 degrees towards `OPEN_ANGLE`.
Some balls stay behind	Increase `HOLD_OPEN_MS`, for example to `2500`.
Arm is roughly right but a bit off	Fine-tune the angle a few degrees at a time, or re-fit the arm one notch over.

Final settings: Closed _____ Open _____ Hold time _____ ms

PART 10

TEST FROM THE STAFF PAGE

10 minutes

1. **On the phone staff will use on the day,** sign in to the staff page and open the `Score` tab.
2. **Check the Ball release panel** shows `• Dropper online` and the Wi-Fi strength. "Strong" and "OK" are both fine.
3. **Load a few balls** behind the closed gate.
4. **Press** `⬇ Release balls`, then `OK`.
5. **Watch the panel and the gate:** the panel shows `• Releasing... waiting for the dropper`, the gate opens and closes, and then the panel shows `• Released ✓ Match N`.
6. **Repeat 5 times** with real balls, including from a second staff phone.
7. **Test a Wi-Fi drop:** turn the hotspot off. Within about 15 seconds the panel shows `• Dropper offline`. Turn it back on: within about 30 seconds it's back to `• Dropper online`.

WHAT THE PANEL CAN SHOW

Panel	Meaning and what to do
• Dropper online	Ready. The ESP32 checked in within the last 10 seconds.
• Releasing... waiting for the dropper	The press was received. The button is locked until the dropper confirms, usually 2–3 seconds.
• Released ✓ Match N	The gate opened and closed.
• Not confirmed by the dropper	No confirmation within 20 seconds. Check power, the LED and the hotspot, then press again.
• Dropper offline	The ESP32 isn't connected: no power, no Wi-Fi, or the hotspot is off.
• Not set up yet	The database step (Part 6) is missing.

BUILT-IN SAFETY

- **No surprise drops:** after a power cut or restart, the ESP32 never repeats an old release.
- **No late drops:** it ignores any release more than 20 seconds old, for example if the Wi-Fi dropped for a while.
- **No double drops:** two presses within 3 seconds, even from two phones, count as one.
- **A record of every release:** each one is logged with the match number and the staff member who pressed it.

PART 11

INSTALL IT ON THE FIELD

30 minutes
day before the event

1. **Mount the servo firmly** with screws, cable ties or strong double-sided tape, so it can't twist when the gate opens. Test again with the test wire after mounting.
2. **Put the ESP32 and breadboard in a small plastic box**, not a metal one, which blocks Wi-Fi. Leave a gap or hole for the wires. Keep it away from the robots, the balls and people's feet.
3. **Secure every wire.** Tape the jumper wires to the breadboard and box so a tug can't pull them out. Loose jumpers are the most common failure at events.
4. **Power for the ESP32:** a phone wall charger is best. If you use a power bank, charge it fully and test it for 30 minutes. Some power banks switch off by themselves when the load is small.
5. **Hotspot phone:** keep it within about 10 m of the box, on a charger, with auto-off disabled. If the panel shows weak Wi-Fi, move the phone closer.
6. **Batteries:** put fresh AAs in on the morning of the event and keep a spare set in the box.
7. **Label the box** "Ball release – please don't touch".
8. **Full rehearsal:** run a complete fake match from the staff phone: start the clock, release the balls, and check `● Released ✓` appears.

PART 12

EVENT-DAY CHECKLIST

Saturday 14 November

30 MINUTES BEFORE DOORS OPEN

- ☐ Hotspot on, on 2.4 GHz (Maximise Compatibility), phone charging
- ☐ Fresh AA batteries in, servo battery switched on
- ☐ ESP32 plugged in, and the blue LED on within about 10 seconds
- ☐ Staff page shows **Dropper online** with Strong or OK Wi-Fi
- ☐ Empty test release from the staff page: the gate opens and closes, and the panel shows **Released ✓**
- ☐ All wires taped down, box closed and labelled
- ☐ Balls loaded with the gate closed

EVERY MATCH

- ☐ Correct match on the display
- ☐ At the start signal, press **⬇ Release balls** and confirm
- ☐ The panel shows **Released ✓ Match N**
- ☐ Reload the balls for the next match

IF SOMETHING GOES WRONG MID-EVENT

1. **Don't hold up the match.** Use the backup: touch the test wire from pin 0 to GND, or drop the balls by hand.
2. **Look at the blue LED.** If it's off, check the ESP32's power and the hotspot.
3. **Servo doesn't move but the panel shows Released ✓?** Check the servo battery and its wires.
4. **Still stuck?** Unplug the ESP32's USB for 5 seconds, then plug it back in. It reconnects within about 10 seconds and won't drop anything by itself.

AFTER THE EVENT

- ☐ Unplug the ESP32 and take the batteries out
- ☐ Store the board in its anti-static bag

PART 13

TROUBLESHOOTING

Serial Monitor at **115200 baud**

You see	Do this
No Windows sound when plugged in	Try another cable and another USB socket. Check that the board isn't sitting on anything metal.
No COM port in **Tools** › **Port**	Use a data cable, not a charge-only one, and install the CP210x or CH340 driver (Part 5).
"esp32" doesn't appear in the Boards Manager	Add the board URL in **File** › **Preferences** (Part 4B).
Upload stuck on `Connecting.....` or "Failed to connect"	Connect pin 0 to GND, press RESET, upload, then remove the wire (Part 7C). Close any other program using the COM port.
`secrets.h is missing` when compiling	Create the secrets.h tab (Part 7B).
Red errors mentioning `secrets.h` or `expected …`	A quote mark `"` is missing or extra in secrets.h. Compare it with the example.
`Wi-Fi not found yet (it must be 2.4 GHz)`	Check the name and password, including capitals. Make the hotspot 2.4 GHz (Part 7B) and move it closer.
`Connection error` repeating	The hotspot has no mobile data or signal. Open a website on the hotspot phone to check.
`Supabase error 404`	The database step is missing. Run `field_release.sql` (Part 6).
`Supabase error 401` or `403`	The token in secrets.h is wrong. Copy it again (Part 6), then upload again.
`Brownout detector was triggered`, or the board restarts when the servo moves	The servo is getting power from the ESP32. It must use the battery pack, with only GND shared (Part 8).
Servo twitches or doesn't move at all	Check wire 7 (GND to the – rail), that pin 18 and the servo signal are in the same column, and the batteries.
Board does nothing after power-up, and the Serial Monitor shows `waiting for download`	Pin 0 was connected to GND at power-up. Pull the test wire off GND and press **RESET**.
Panel: **Not set up yet**	Run `field_release.sql` (Part 6), then reload the staff page.
Panel: **Dropper offline** while the Serial Monitor says it's connected	Reload the staff page. Check the staff phone has internet.
Panel: **Not confirmed**, but the gate did open	The confirmation was lost because the Wi-Fi dropped right after. The balls were released, so you don't need to press again.

MAKE A NEW DEVICE TOKEN

Do this if the token was shared by accident. Run these two lines in the Supabase SQL Editor, put the new token in secrets.h, and upload again:

```
update public.field_devices set token = replace(gen_random_uuid()::text, '-', '') where name = 'dropper';
select token from public.field_devices where name = 'dropper';
```

PART 14

QUICK REFERENCE CARD

Print this page and keep it in the box with the ESP32

WIRING

Servo signal	→ ESP32 pin **18** (same breadboard column)
Servo red	→ battery + (breadboard + rail)
Servo brown	→ battery – (– rail)
ESP32 GND	→ – rail **(required)**
ESP32 power	USB only

BOARD LIGHTS AND BUTTONS

CHG LED on or flickering	Has power (normal)
Blue LED on	Connected, ready to release
Blue LED off	No Wi-Fi or no connection
Pin **0** to **GND** (a moment)	Test release, works offline
RESET	Restart (never drops by itself)

STAFF PAGE PANEL

Dropper online	Ready
Releasing...	Waiting about 2–3 s
Released ✓	Done
Not confirmed	Check power and Wi-Fi, press again
Dropper offline	No power or no Wi-Fi

SETTINGS ON THIS BOARD

Wi-Fi (hotspot)	_____
Backup Wi-Fi	_____
COM port (setup PC)	_____
Closed / Open angle	_____ / _____
Hold time	_____ ms

CHANGING SOMETHING LATER

1. Plug the ESP32 into the setup PC and open `ball_release.ino` in Arduino IDE.
2. Change the **secrets.h** tab for a new Wi-Fi or token, or the angle lines at the top of **ball_release.ino**.
3. Press Ctrl + S, check the board and COM port, then click **Upload** →. If it gets stuck on Connecting, see Part 7C.
4. Check the Serial Monitor (115200 baud) says `Connected to Supabase`.

Backup plans at the field, in order: touch the test wire from pin **0** to **GND** → unplug and replug the ESP32's USB → drop the balls by hand. Never hold up a match for the dropper.